

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

Stefano Covino
INAF / Brera Astronomical Observatory



Statistics Reminder

Probability

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- We adopt here a Bayesian view: probability is just a degree of certainty about a statement.
- We describe probability following the so-called Bayesian system of probability.
- An experiment is any action that can have a set of possible results where the actually occurring result cannot be predicted with certainty prior to the action.

The set Ω of all outcomes of an experiment is known as the outcome space or sample space.

A well-balanced coin toss gives $\Omega = H, T$, and the inherent symmetries of the experiment leads to $P(H) = P(T) = 0.5$.

Axioms of probability

- A probability space consists of the triplet $\Omega, \mathcal{F}, P$, a sample space, a class of events, and a function that assigns a probability to each event in $\mathcal{F}$ following:
 - Axiom 1: $0 \leq P(A) \leq 1$, for all events A
 - Axiom 2: $P(\Omega) = 1$
 - Axiom 3: For mutually exclusive (pairwise disjoint) events $A_1, A_2, \dots$,

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

- or, if for any $i \neq j, A_i \cap A_j = \emptyset$, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

- From these, we have that for generic events, not necessarily disjoint, C and D, we have:

$$P(C \cup D) = P(C) + P(D) - P(C \cap D)$$

- A formal proof of the previous statement can be obtained observing that:

$$\begin{aligned} P(C \cup D) &= P(C \cap D^c) + P(C \cap D) + P(D \cap C^c) = \\ &= P(C \cap D^c) + P(C \cap D) + P(D \cap C^c) + P(C \cap D) - P(C \cap D) = \\ &= P(C \cap (D \cup D^c)) + P(D \cap (C \cup C^c)) - P(C \cap D) = \\ &= P(C \cap \Omega) + P(\Omega \cap D) - P(C \cap D) = P(C) + P(D) - P(C \cap D) \end{aligned}$$

Conditional probability

- This is a far from trivial concept in probability:

$$P(A|B) \equiv \frac{P(A \cap B)}{P(B)}$$

- The formula above can be read as “probability of A given B (i.e. knowing that B occurred)”.

E.g., rolling a dice, and with $A = 1, 2, 3$, implies $P(A) = 1/2$. The probability of an even outcome ($B = 2, 4, 6$) is again $1/2$, but $P(A \cap B)$ is $1/6$. Then, $P(A|B) = 1/3$.

Bayes' Theorem

- Introduced by Thomas Bayes, but recognized earlier by James Bernoulli and Abraham de Moivre, and later fully explicated by Pierre Simon de Laplace.

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

- $P(A|B)$: the **posterior**, or the probability of the model parameters given the data: this is the result we want to compute.
- $P(B|A)$: the **likelihood**, the same function, but with a different meaning, used in the frequentist approach.
- $P(A)$: the **model prior**, which encodes what we knew about the model prior to the application of the data.
- $P(B)$: the data probability or **evidence**, in most cases just a normalization factor, but fundamental for model inference.

It is intriguing that most of the debate between “frequentists” vs “Bayesians” is not due to the mathematics of the theorem, but to its philosophical meaning, i.e. the basis of Bayesian inference.

Before discussing Bayesian inference, we now recall some useful “frequentist” concepts and algorithms.

Basic definitions and some useful “frequentist” tools

- *Random variable*: variable describing the possible output of an experiment.
 - It can be discrete or continuous.
- *Parent distribution*: distribution of values for a RV if the experiment is repeated number of times.

These definitions are somehow reconsidered in a Bayesian scenario, but for now let's avoid these, interestingly actually, issues,

- *Mean* of parent distribution: $\mu \equiv \lim_{N \rightarrow \infty} \left(\frac{1}{N} \sum_i x_i \right)$
- *Median* of parent distribution: the median value, $\mu_{1/2}$ is where 50% of the input data are below and above the identified value.
 - Computing the median requires to sort the data.
- *Mode* (or most probable value): $P(\mu_{max}) \geq P(x \neq \mu_{max})$.

For symmetric distributions the mean, media and mode coincide.

- Standard deviation (σ) and variance (σ^2) of the parent distribution:

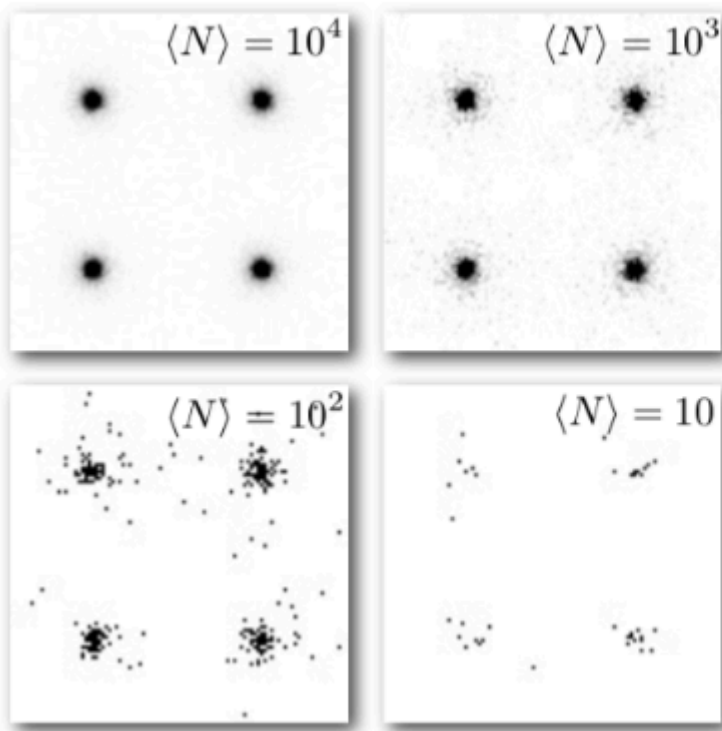
$$\sigma^2 \equiv \lim_{N \rightarrow \infty} \left[\frac{1}{N} \sum_i (x_i - \mu)^2 \right]$$

- And standard deviation (s) and variance (s^2) of a sample population:

$$s^2 = \frac{1}{N-1} \sum_i (x_i - \mu)^2$$

Counting statistics

- With modern digital devices a large fraction of what we call "measurement" consists in counting discrete events (photons, etc.)



Photon counting is a *random process*.

Repeated measurements will not yield identical results but be subject to random fluctuations.

How to quantify this?

- Let's assume to count an average number of events μ in a given time interval t .
- And divide our time interval in a large number of sub-intervals n , so that the probability to detect one event in one sub-interval becomes $p \approx \mu/n$.
- Then, the probability to detect k events in our time interval is given by the Binomial probability distribution:

$$P(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1 - p)^{n-k}$$

or

$$P(k; \mu) = \frac{n!}{k!(n-k)!} \left(\frac{\mu}{n}\right)^k \left(1 - \frac{\mu}{n}\right)^{n-k}$$

that, in the limit of large n , very low probability, becomes the Poisson distribution:

$$P(k; \mu) = \frac{\mu^k}{k!} e^{-\mu}$$

Binomial distribution

- Applicable when carrying out a well defined number of trials, n , each of which has probability p of success.
 - Example: rolling $n = 100$ dice, which is the probability of getting exactly **15** sixes ($p = 1/6$)? $P(15, 100, 1/6) \approx 10\%$.

Number of trials:

Probability of success of a single trial:

Number of successes:

Probability of successes: **9.60%**

Note that this is the probability of obtaining exactly the required number of successes!

- The probability of obtaining the required number or less (i.e. up to the required number) can also be easily computed.

Probability of at least 15 successes: **35.42%**

Poisson distribution

- Applicable when the mean number of outcomes, μ , is known or can be estimated and probability $p \ll 1$.
 - Example: About 1% of pregnancies are twin pregnancies. In **1000** pregnancies, which is the probability of having exactly **5** twin pregnancies? $P(5, 10) \approx 4\%$.

Number of trials:

Probability of success of a single trial:

Number of successes:

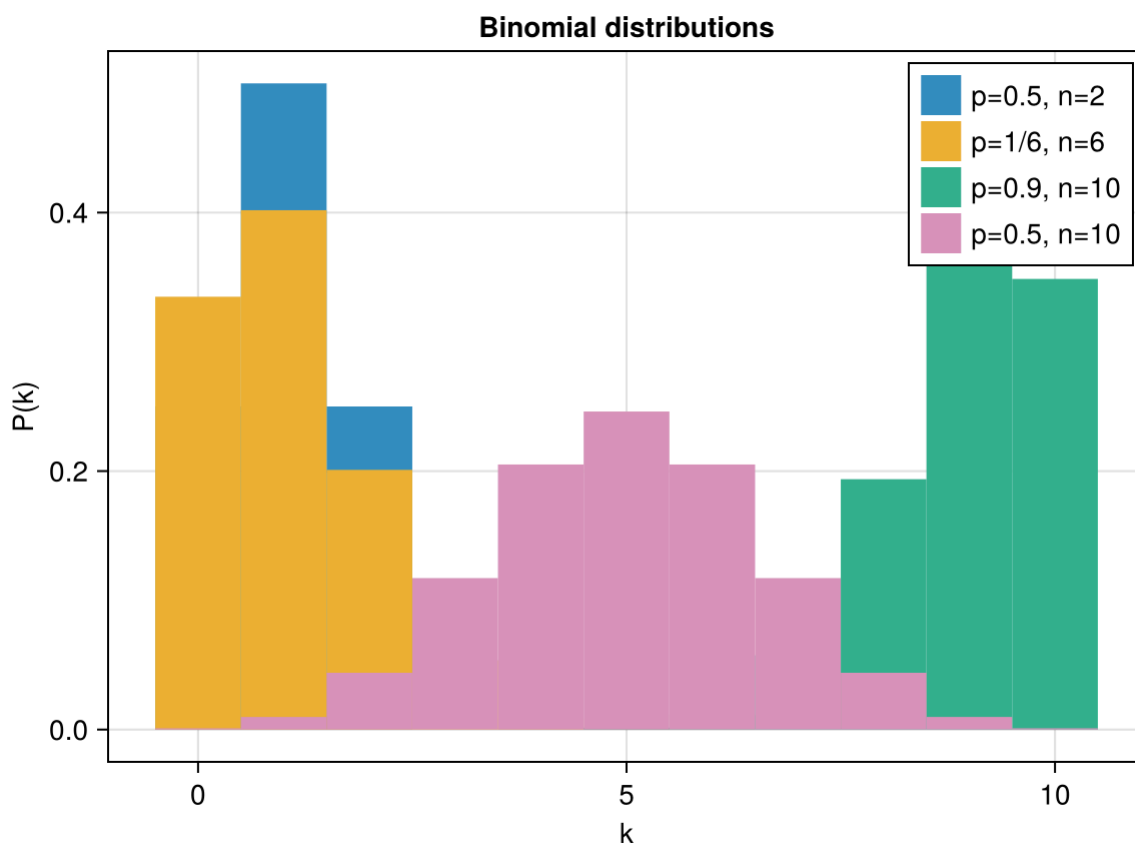
Probability of successes: **3.78%**

Note again that this is the probability of obtaining exactly the required number of successes!

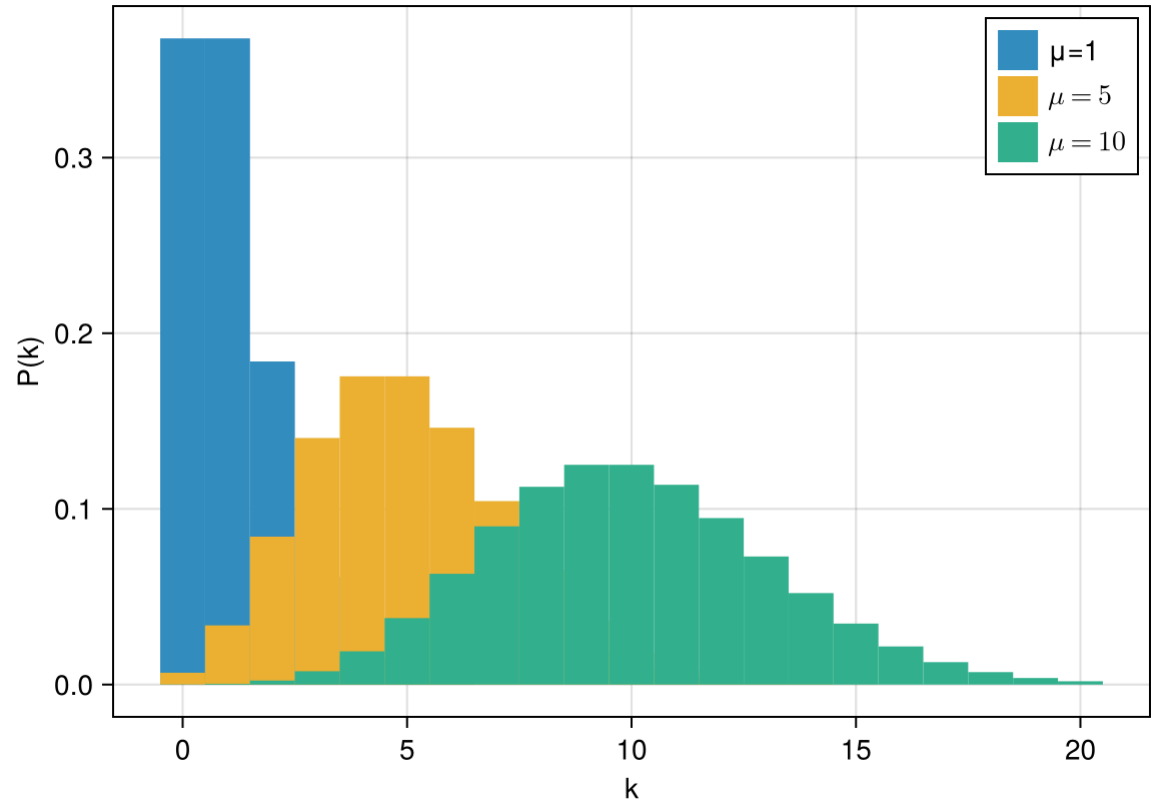
- The probability of having up to the required number of twin pregnancies can instead be computed as:

Probability of up to 5 twin pregnancies: **6.71%**

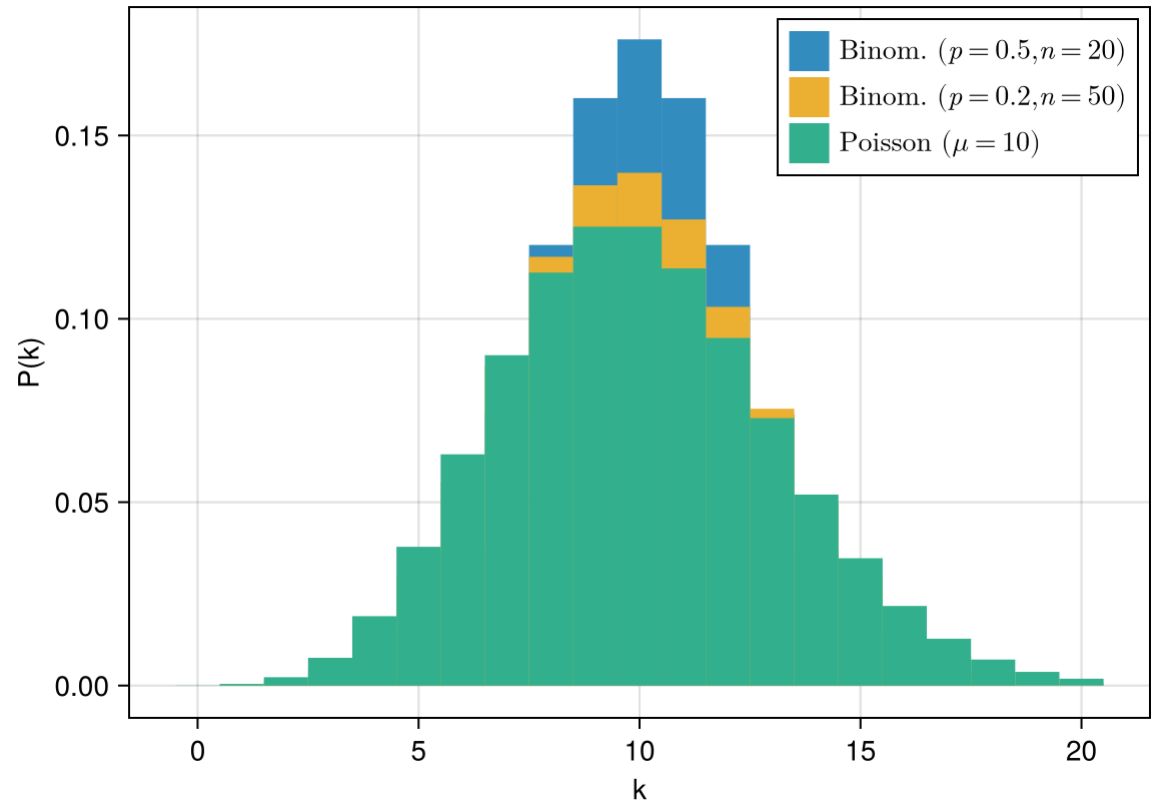
- Let's see now a few plots about the Binomial and Poisson distributions:



Poisson distributions



Poisson distributions



Mean and variance

=====

- For any probability density function of a discrete variable $P(x_i)$:

- Mean: $\mu = \sum_{i=0}^{\infty} x_i P(x_i)$,
- Variance: $\sigma^2 = \sum_{i=0}^{\infty} (x_i - \mu)^2 P(x_i)$

- For a binomial distribution:

$$\mu = \sum_0^n k P(k; n, p) = \sum_0^n k \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = np$$

- In fact:

$$\begin{aligned} \mu &= \sum_1^n k \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = \sum_1^n \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} = \\ &= \sum_1^n \frac{n(n-1)!}{(k-1)!(n-k)!} p p^{k-1} (1-p)^{n-k} = np \sum_1^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} (1-p)^{n-k} \end{aligned}$$

- Then: $\mu = np \sum_1^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} (1-p)^{n-k} = np \sum_1^n k \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k}$
- Let's rename $n-1 \equiv m$ and $k-1 \equiv j$ so that $\mu = np \sum_j^m \binom{m}{j} p^j (1-p)^{m-j}$.
- The summed term is just 1, i.e. the sum of all the possible outcomes of a binomial distribution.
Thus: $\mu = np$.

$$\sigma^2 = \sum_0^n (k - np)^2 \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = np(1-p)$$

- In fact, $\sigma^2 = \mathbb{E}[k^2] - \mathbb{E}[k]^2 = \mathbb{E}[k^2] - [np]^2$ and we just need to compute the first term.
- As above, we can write the first term as: $\mu = \sum_1^n k^2 \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$, we now write $k^2 = k(k-1) + k$, so that we have:

$$\sum_1^n k^2 \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = \sum_1^n k(k-1) \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} + \sum_1^n k \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

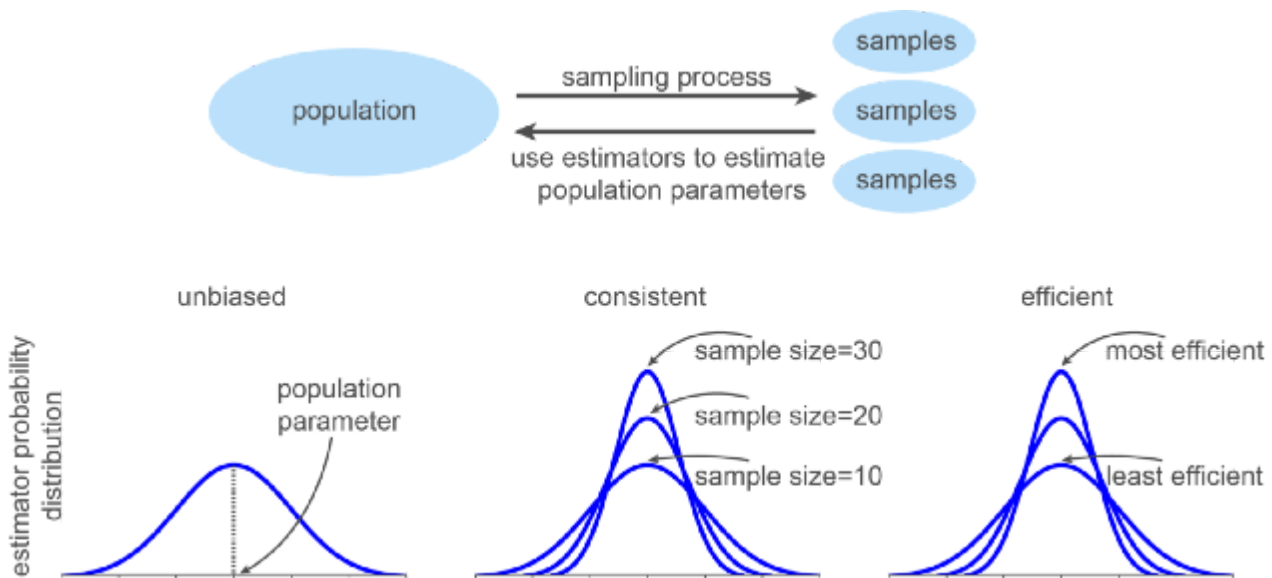
- The last term is again np and, again following the same procedure we applied before, we have that: $\sum_1^n k(k-1) \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = n(n-1)p^2 \sum_2^n \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} (1-p)^{n-k}$
- Following the procedure we applied for the *mean* above we, finally, have:
 $\sigma^2 = n(n-1)p^2 + np - (np)^2 = -np^2 + np = np(1-p)$

In the limit of large n and small p , i.e. Poisson distribution, $\sigma^2 = np = \mu$.

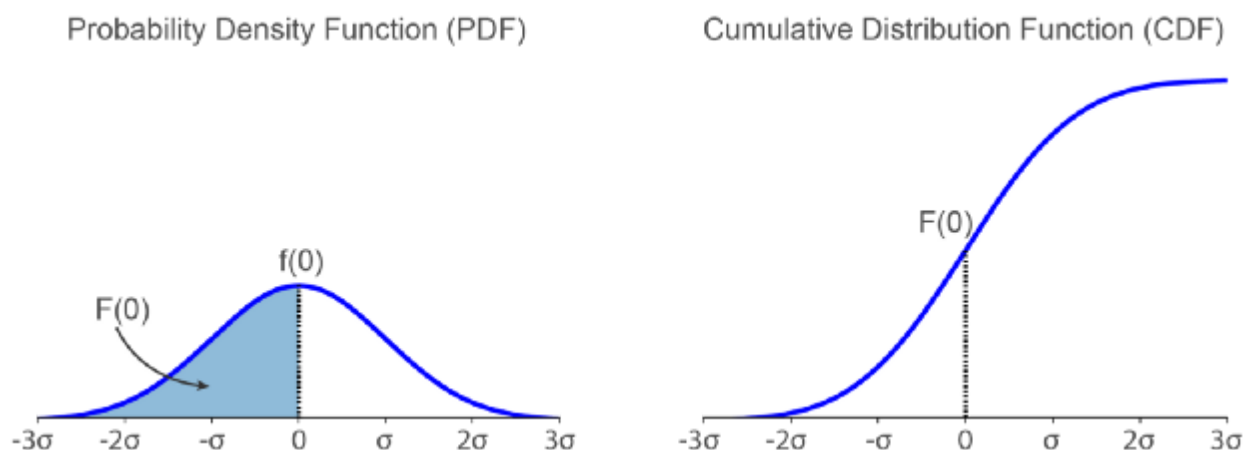
Error on the mean

- Mean of N measurements $x_1, x_2, \dots, x_N$: $\bar{x} = \frac{1}{N} \sum_i x_i$
- Propagation of error on the mean:

$$\sigma_{\bar{x}}^2 = \left(\frac{\partial \bar{x}}{\partial x_1}\right)^2 \sigma_{x_1}^2 + \left(\frac{\partial \bar{x}}{\partial x_2}\right)^2 \sigma_{x_2}^2 + \dots = \left(\frac{1}{N}\right)^2 \sigma_{x_1}^2 + \left(\frac{1}{N}\right)^2 \sigma_{x_2}^2 + \dots = \frac{1}{N^2} \sum_i \sigma_{x_i}^2$$
 - If $\sigma_{x_1} = \sigma_{x_2} = \dots = \sigma_i$, then $\sigma_{\bar{x}}^2 = \frac{N\sigma_i^2}{N^2} = \frac{\sigma_i^2}{N} \Rightarrow \sigma_{\bar{x}} = \frac{\sigma_i}{\sqrt{N}}$
- Speaking of estimators, there are three properties to define a good estimator: unbiased, consistent, and efficient.
 - The estimator is unbiased, when the expected value of the sample parameter is equal to the population parameter.
 - If the variance of the sample parameter decreases with the increasing sample size, the estimator is consistent.
 - With the same sample size, the estimator with lower variance is more efficient.



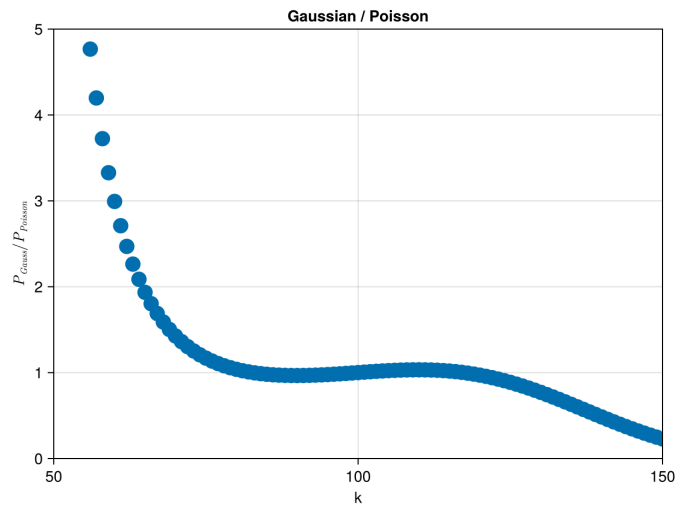
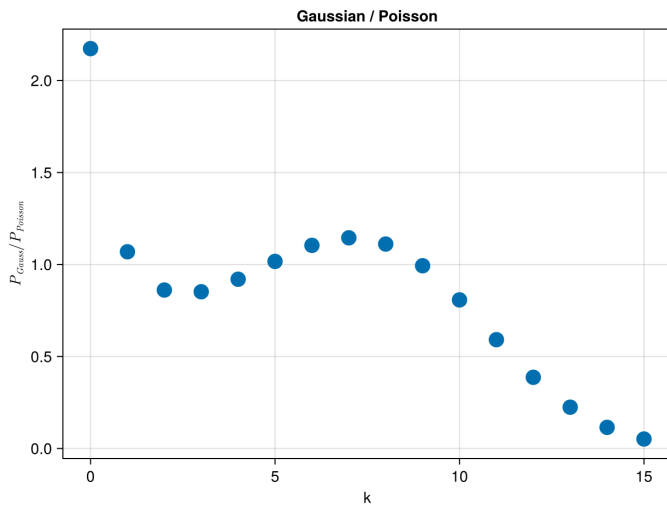
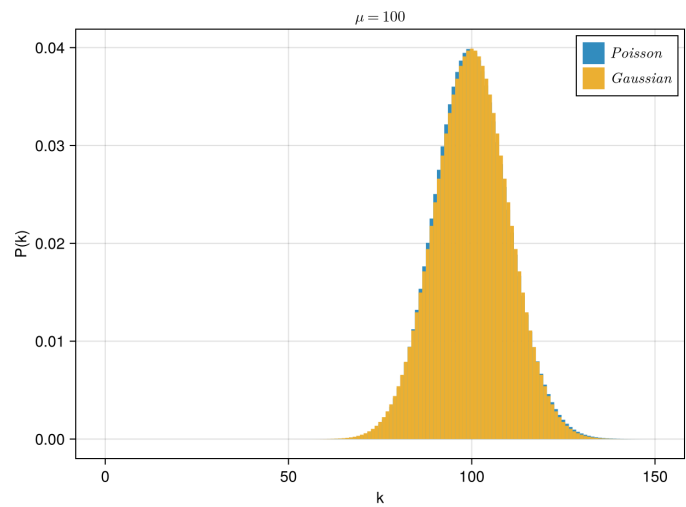
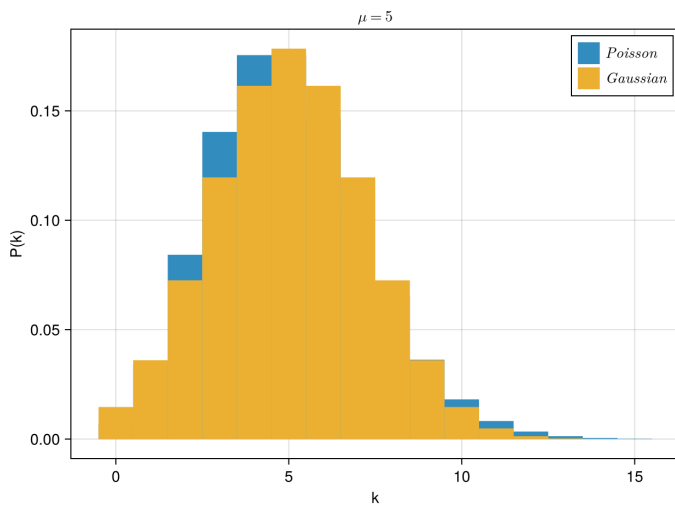
- By a Probability Density Function the probability of the random variable falling within a particular range of values can be calculated.
 - If the probability density at a certain x is denoted as $f(x)$. By applying integral function to $f(x)$ over a range of (x_1, x_2) , the probability of x falling in (x_1, x_2) can be calculated:



The Normal distribution

- For large μ the Poisson distribution can be approximated by a Gaussian (with $\mu = \sigma^2$).

$$P_G(x; \sigma, \mu) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$



Inverse distribution

- An interesting problem, having to deal with probability distributions, is how to compute the *inverse distribution*, i.e. the distribution of the reciprocal of a random variable.
- In general, given the probability distribution of a random variable X with strictly positive support, it is possible to find the distribution of the reciprocal, $Y = 1/X$.
 - If the distribution of X is continuous with density function $f(x)$ and cumulative distribution function $F(x)$, then the cumulative distribution function, $G(y)$, of the reciprocal is found by:

$$G(y) = \Pr(Y \leq y) = \Pr(X \geq \frac{1}{y}) = 1 - \Pr(X < \frac{1}{y}) = 1 - F(\frac{1}{y})$$

- Therefore, the density function of Y is found as the derivative of the cumulative distribution function:

$$g(y) = \frac{1}{y^2} f(\frac{1}{y})$$

Exercise: the inverse of the uniform distribution

- If $\mathbf{X}$ is uniformly distributed in a given interval (a, b) , with $a > 0$, $f(x) = \frac{1}{b-a}$.
- The distribution of the reciprocal $\mathbf{Y} = 1/\mathbf{X}$ takes values in the range (b^{-1}, a^{-1}) , and turns out to be:

$$g(y) = y^{-2} \frac{1}{b-a}$$

The inverse CDF method

- We have introduced before the uniform random variable, i.e., a continuous random variable which takes on values from parameters a to b .
- The continuous probability distribution of a uniform random variable X is: $f(x) = \frac{1}{b-a}$. If $a = 0$ and $b = 1$ we have the so-called standard uniform and $f(x) = 1$.
- A cumulative distribution function (CDF) is the probability that a real-valued random variable X with a given probability distribution is less than or equal to a quantity x . It is often denoted by $F(x) = P(X \leq x)$.
- Some of the CDF properties are:
 1. The CDF is a non-decreasing function.
 2.
$$\lim_{x \rightarrow +\infty} F(x) = 1$$
 3.
$$\lim_{x \rightarrow -\infty} F(x) = 0$$
- The CDF of a uniform random variable X in the interval $[a, b]$ is $F(x) = \frac{x-a}{b-a}$ since $F(x) = P(X \leq x) = \int_a^x \frac{1}{b-a} dx = \frac{x-a}{b-a}$.
- The CDF of a standard random variable is just $F(x) = x$.
- The *CDF method* allows one to generate any non-uniform random variables with known CDF starting from the uniform random case:
 1. Obtain or generate a draw (realization) u from a standard uniform distribution.
 2. The draw x from CDF $F(x)$ is given by $x = F^{-1}(u)$.

Exercise: A uniform distribution in the (a, b) interval

- This is definitely a contrived example. Yet, the CDF of the distribution we want to sample is, as we know: $F(x) = \frac{x-a}{b-a}$.
- Let's solve for x in $F(x) = u$. We get $F^{-1}(u) = a + (b-a)u$. If we sample from $\mathcal{U} \sim (0, 1)$ we get samples for X .

Exercise: Samples from the exponential distribution

- We want to generate an exponential random variable with $x > 0$ and the rate lambda as $\lambda > 0$.
The continuous probability distribution of the exponential random variable is: $f(x) = \lambda e^{-\lambda x}$.
- Integrating $f(x)$ from 0 we have: $F(x) = \int_0^x \lambda e^{-\lambda u} du = 1 - e^{-\lambda x}$.
- Solving for $x = F^{-1}(u)$ gives $x = -\frac{1}{\lambda} \ln(1 - u)$.

Exercise: Samples from the Pareto distribution

- Now, we deal with the Pareto distribution. Given the shape parameter k and the scale parameter λ , the Pareto distribution has PDF: $f(x) = \frac{k\lambda^k}{x^{(k+1)}}$.
- The CDF of the Pareto distribution, integrating from λ to x is: $F(x) = P(X \leq x) = \int_{\lambda}^x \frac{k\lambda^k}{t^{(k+1)}} dt$.
- Solving the integral we have:

$$F(x) = \frac{k\lambda^k t^{-k}}{-k} \Big|_{t=\lambda}^{t=x} = \frac{k\lambda^k x^{-k}}{-k} - \frac{k\lambda^k \lambda^{-k}}{-k} = -\lambda^k x^{-k} + 1 = 1 - \left(\frac{\lambda}{x}\right)^k.$$
- Now, let's solve for $F(x) = u = 1 - \left(\frac{\lambda}{x}\right)^k$ and we get: $x = F^{-1}(u) = \frac{\lambda}{(1-u)^{1/k}}$.

Central limit theorem

One of the most important results in statistics

- No matter what distribution (with finite mean and variance) our population follows, as we increase sample size, sampling distribution of the mean converges to a Normal distribution.
- One can also say that the convolution of a large number of (positive) functions $f_i(x)$ with variances σ_i^2 converges to a Gaussian with $\sigma^2 = \sum \sigma_i^2$.
 - The functions do not need to be Gaussian.

Just pay attention since the CLT is more subtle than it seems! For interested readers a verbose but sufficiently clear proof is discussed here ([notebook](#), [html](#)).

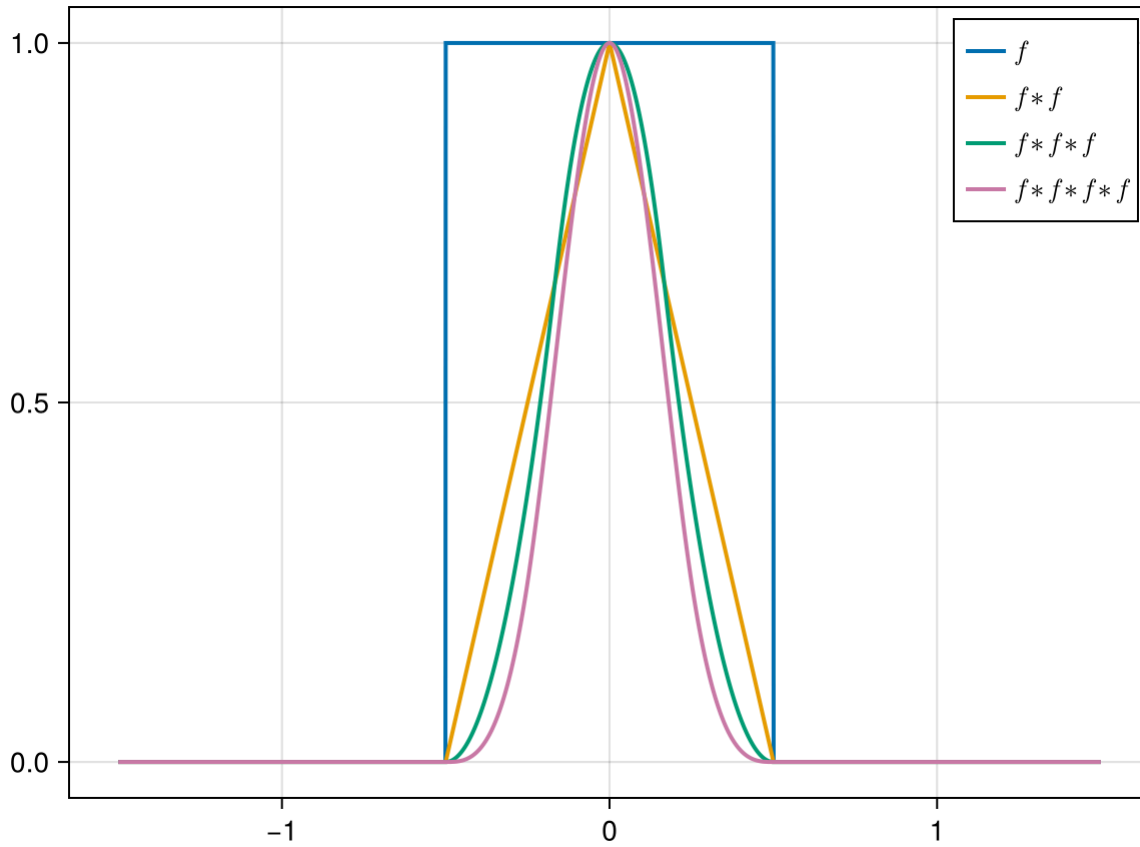
Convolution

- Mathematically, the convolution of two functions $f(x)$ and $g(x)$ is defined as:

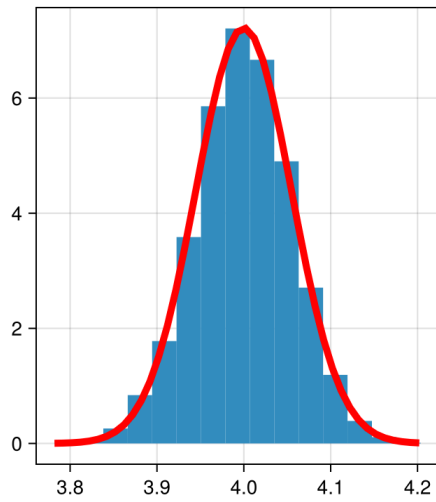
$$(f * g)(x) = \int_{-\infty}^{\infty} f(\xi)g(t - \xi)d\xi$$

- That, as we will see further in the course, it is conveniently carried out in the Fourier space:

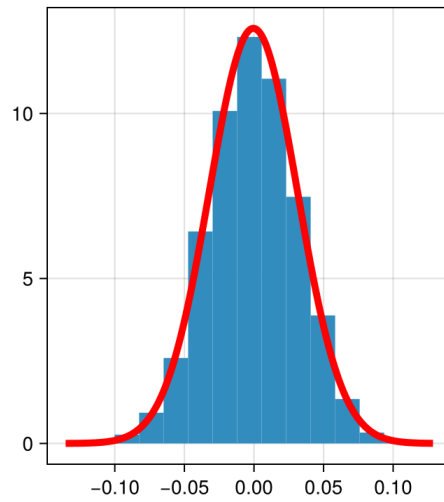
$$\mathcal{F}\{f * g\} = \mathcal{F}\{f\} \cdot \mathcal{F}\{g\}$$



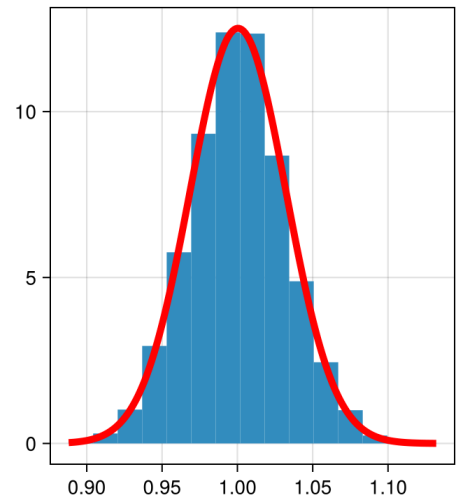
Uniform distribution



Normal distribution



Poisson distribution



```

1  begin
2      unimeanList = []
3      normeanList = []
4      poimeanList = []
5
6
7      num_trials = 10000
8      num_observations = 1000
9
10
11     unidistr = Uniform(1,7)
12     nordistr = Normal(0,1)
13     poidistr = Poisson(1)
14
15     for i in 1:num_trials
16         # sample from uniform distribution
17         uninumList = rand(unidistr,num_observations)
18         # sample from normal distribution
19         nornumList = rand(nordistr,num_observations)
20         # sample from poisson distribution
21         poinumList = rand(poidistr,num_observations)
22         #
23         push!(unimeanList,mean(uninumList))
24         push!(normeanList,mean(nornumList))
25         push!(poimeanList,mean(poinumList))
26     end
27
28
29     unifit = fit(Normal, Float64.(unimeanList))
30     norfit = fit(Normal, Float64.(normeanList))
31     poifit = fit(Normal, Float64.(poimeanList))
32
33     #
34
35
36     fg5 = Figure(size=(1000,400))
37
38     ax1fg5 = Axis(fg5[1, 1],
39         title="Uniform distribution"
40     )
41
42     hist!(unimeanList,normalization=:pdf)
43     rng = range(start=minimum(unimeanList),stop=maximum(unimeanList),step=0.01)
44     lines!(rng,pdf(unifit,rng),color=:red,linewidth=5)
45
46     ax1fg5 = Axis(fg5[1, 2],

```

```

47     title="Normal distribution"
48 )
49
50 hist!(normeanList,normalization=:pdf)
51 rng = range(start=minimum(normeanList),stop=maximum(normeanList),step=0.001)
52 lines!(rng,pdf(norfit,rng),color=:red,linewidth=5)
53
54 ax1fg5 = Axis(fg5[1, 3],
55     title="Poisson distribution"
56 )
57
58 hist!(poimeanList,normalization=:pdf)
59 rng = range(start=minimum(poimeanList),stop=maximum(poimeanList),step=0.001)
60 lines!(rng,pdf(poifit,rng),color=:red,linewidth=5)
61
62 fg5
63
64 end

```

Maximum Likelihood Estimation

- For a given set of observations $\mathbf{x}_i \{i = 1 \dots N\}$, and assuming a parent distribution $P(\mathbf{x}; \xi_1, \xi_2, \dots)$, let's find the set of parameters $\xi_1, \xi_2, \dots$ so that the probability to observe the values $\mathbf{x}_i$ is maximized.
- Let's assume a Gaussian parent distribution:

$$P_G(\mathbf{x}; \sigma, \mu) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- The total likelihood to observe the whole dataset turns out to be (with the, often hidden, assumption of statistical independence of the data):

$$\mathcal{L}(\mu, \sigma; \mathbf{x}_i) = \prod_i P_G(\mathbf{x}_i; \sigma, \mu) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^N e^{(-\frac{1}{2} \sum_i \frac{(x_i - \mu)^2}{\sigma^2})}$$

- It is usually simpler working with the logarithm of the likelihood:

$$\log \mathcal{L}(\mu, \sigma; \mathbf{x}_i) = -\frac{N}{2} \log 2\pi - \frac{N}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2$$

- that is maximized at $\partial P_G(\mu, \sigma) / \partial \mu = 0$ and $\partial P_G(\mu, \sigma) / \partial \sigma^2 = 0$, yielding the maximum likelihood estimators (MLE).
- For this problem, the solution can easily be found analytically:

$$\circ \quad \partial P_G(\mu, \sigma) / \partial \mu \propto \sum_i (x_i - \mu) = 0$$

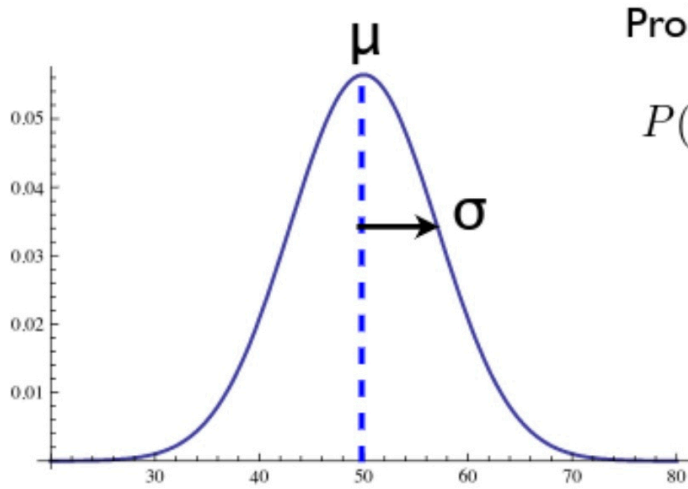
when $\mu = \frac{1}{N} \sum_i x_i$. Not surprisingly, exactly the definition of the sample mean!

$$\begin{aligned} \partial P_G(\mu, \sigma) / \partial \sigma^2 &= -\frac{N}{2\sigma^2} - \frac{1}{2} \sum_i (x_i - \mu)^2 \frac{d}{d\sigma^2} (1/\sigma^2) = -\frac{N}{2\sigma^2} + \frac{1}{2} \sum_i (x_i - \mu)^2 \frac{1}{\sigma^4} = \\ &= \frac{1}{2\sigma^2} \left[\frac{1}{\sigma^2} \sum_i (x_i - \mu)^2 - N \right] \end{aligned}$$

- Then, if $\sigma^2 \neq 0$, can be 0 if $\sigma^2 = \frac{1}{N} \sum_i (x_i - \mu)^2$, i.e. the unadjusted sample variance.
- In general, however, there is no analytical solution and we have to refer to numerical methods to maximize the $\log \mathcal{L}$ or minimize the negative of the previous relation.
- The uncertainty on the parameters can be better estimated by Monte Carlo methods.

Significance level and p-value

- The probability of the observed value is called p-value.
 - A low p value means that the observation is unlikely to occur under the condition that the null hypothesis holds true.
 - When the p value is lower than significance level, then we reject the null hypothesis.



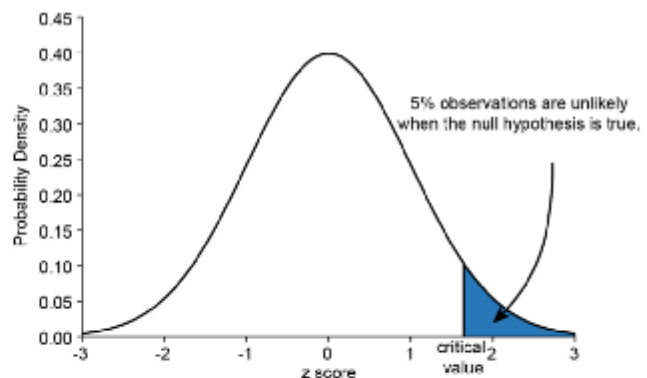
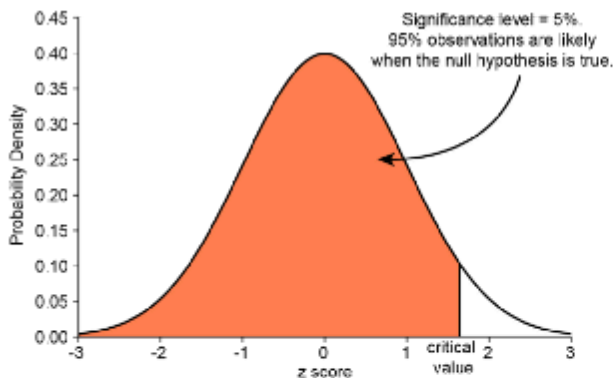
Probability of outcome within 1σ of μ :

$$\begin{aligned}
 P(\mu - \sigma \leq x \leq \mu + \sigma) &= \int_{\mu - \sigma}^{\mu + \sigma} P(x) dx \\
 &= \frac{1}{\sigma \sqrt{2\pi}} \int_{\mu - \sigma}^{\mu + \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx \\
 &= 68.2\%
 \end{aligned}$$

n	1	2	3	5
$P(\mu - n\sigma < x < \mu + n\sigma)$	68.2%	95.5%	99.7%	$1 - 5.7 \times 10^{-7}$

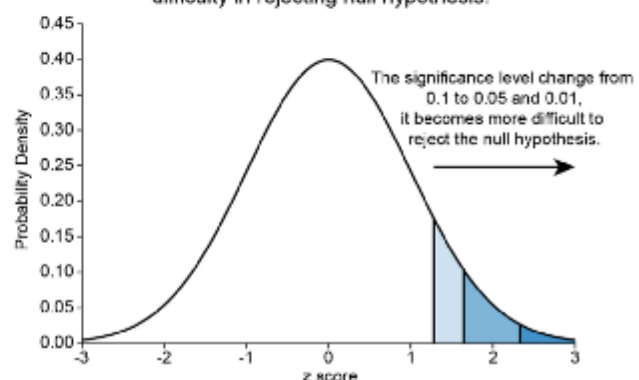
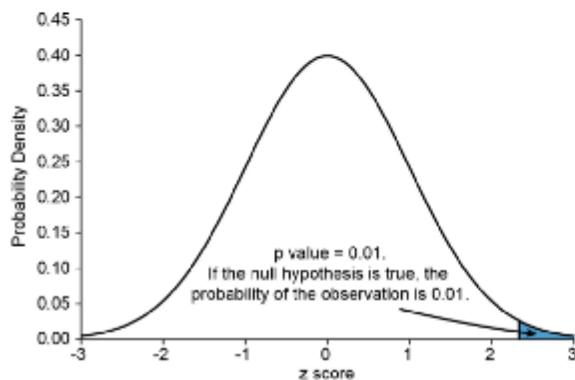
The likely observation, do not reject null hypothesis.

The unlikely observation, reject null hypothesis.



p value

Significance level determines the difficulty in rejecting null hypothesis.



The bootstrap method

- The idea is rather simple. Let's assume to have a dataset $\{x_i\}, i = 1 \dots N$, and define any function $T(x_i)$ of the measurements.
- Be T the value of the function computer for our dataset, and T_j the value of the fuction computed for any nubmer of subsamples built drawing n data from the oiginal dataset with *replacement*, i.e. single value can be picked up multiple times.
- The the variance on T , computed for the original dataset, if we have drawn m subsamples, is:

$$\sigma_T^2 = \frac{1}{m} \sum_{j=1}^m (T_j - T)^2$$

- This is the simplest variant of bootstrap (there are many...), and assumes again statistical independence of the data.

```

1 begin
2   # Let's define a dataset
3   dataset = [16,11,15,16,11,12,18,5,3,1]
4
5   # Let's compute the mean and the median
6
7   μdt = mean(dataset)
8
9   μ½dt = median(dataset)
10
11   #printfmtln("μ = {:.2f} and μ½ = {:.2f}", μdt, μ½dt)
12
13   # Number of resampling
14   n_boot = 1000
15
16   bs1 = bootstrap(mean, dataset, BasicSampling(n_boot))
17   bs2 = bootstrap(median, dataset, BasicSampling(n_boot))
18
19   cil = 0.68;
20
21   ## basic CI
22   bci1 = confint(bs1, NormalConfInt(cil))
23   bci2 = confint(bs2, NormalConfInt(cil))
24
25   #printfmtln("1σ uncertainty for the mean: {:.2f}", (bci1[1][3]-bci1[1][2])/2)
26   #printfmtln("1σ uncertainty for the median: {:.2f}", (bci2[1][3]-bci2[1][2])/2)
27
28 end;
```

μ = 10.80

1σ uncertainty for the mean = 1.74

1σ uncertainty for the median = 2.66

- The uncertainty on the median is always larger than the one on the mean ($\sigma_{1/2} \approx \frac{4}{3}\sigma$).
- However, the median is much less sensitive to outliers.

(Frequentist) hypothesis testing

- In hypothesis testing, a set of complementary hypotheses is proposed, which consists of a null hypothesis and an alternative hypothesis.
- When conducting the hypothesis testing, we choose to believe that the null hypothesis holds true.
 - If the observed value is likely to occur under the condition that the null hypothesis is true, then we do not reject the null hypothesis.
 - However, if the observed value is unlikely to occur, then we reject the null hypothesis and accept the alternative hypothesis.

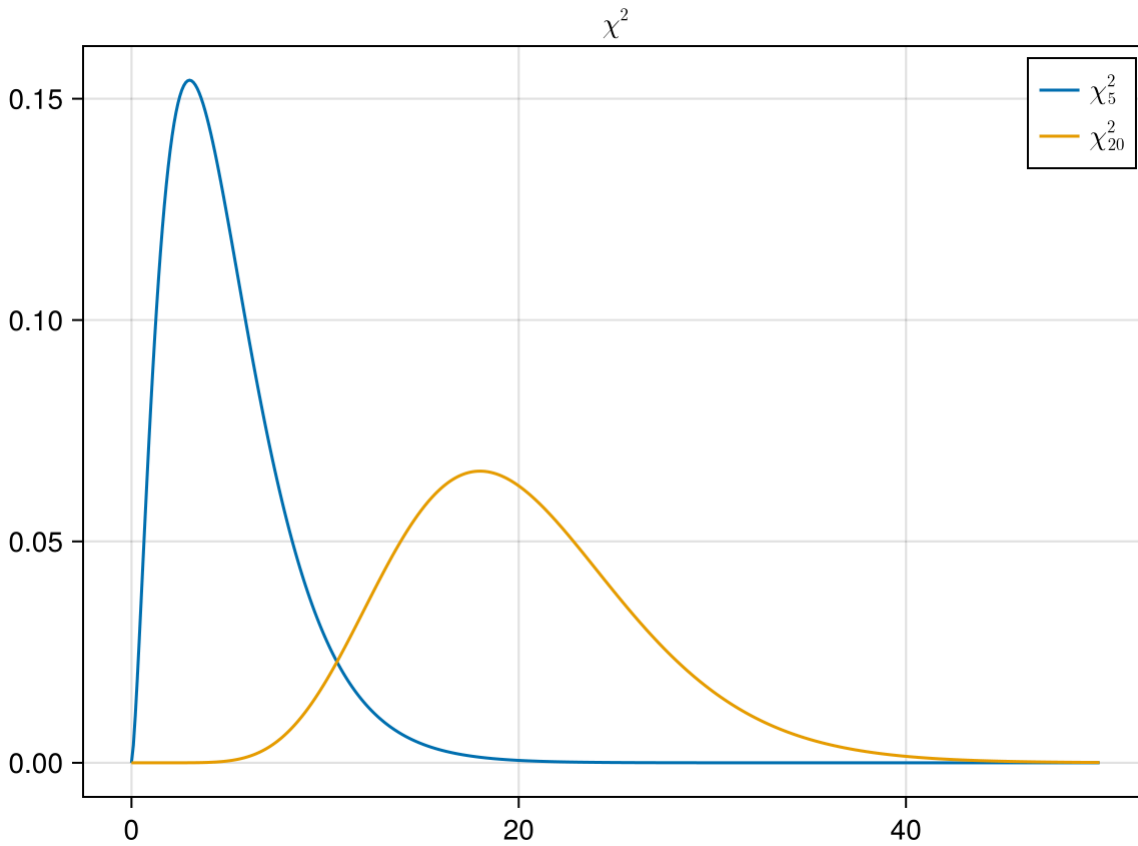
	<i>True negative</i>	<i>False negative</i>
	H_0 True	H_0 False
Don't reject H_0	Correct	Type II error
Reject H_0	Type I error	Correct
	<i>False positive</i>	<i>True positive</i>

- In order to conduct hypothesis testing, we need to define a significance level. Significance level determines the level that we want to believe in the null hypothesis.
 - If we set the significance level as 0.05, then as long as the probability of the observation is higher than 5%, we do not reject the null hypothesis.
 - However, if the probability of the observation falls below 5%, we reject the null hypothesis and accept the alternative hypothesis.
- There is a tradeoff between the Type I and Type II error. Basically, a higher significance level makes it easier to reject the null hypothesis. Although in this way, a higher significance level reduces the Type II error, it also results in a higher Type I error at the same time.
- The only way to reduce both Type I and Type II error is by increasing the sample size...

The χ^2 distribution

- We need to define the χ^2 distribution.
 - If we carry our N measurements $\{x_i\}$ of a random variable $\mathbf{x}$ with variance $\sigma^2 = 1$ and mean $\mu = 0$, then the sum: $\sum_{i=1}^N x_i^2$ follows the χ^2 distribution with N degrees of freedom.

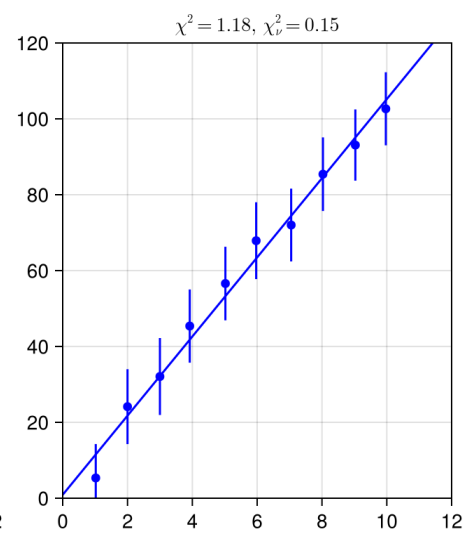
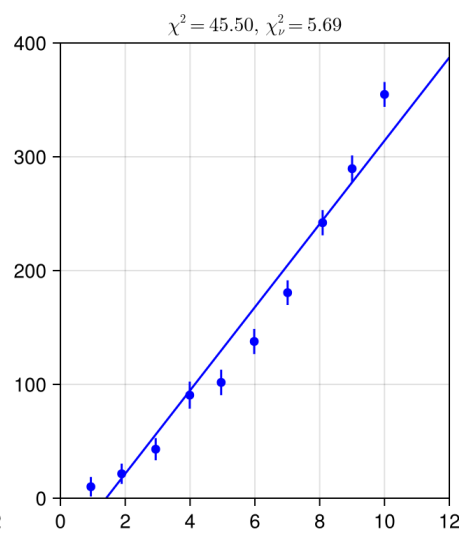
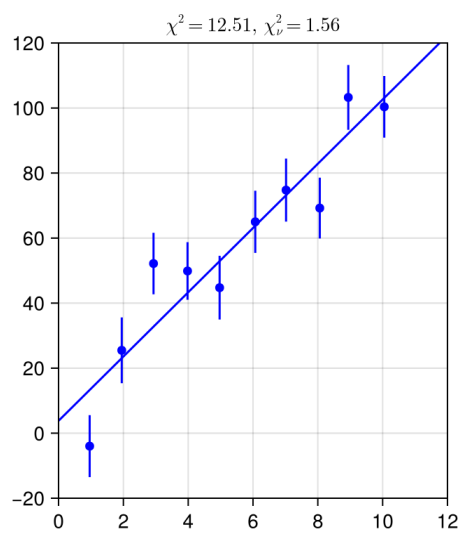
χ^2 degrees of freedom:



- The mean of $\chi^2_N = N$, although often one works with the *reduced* χ^2 , i.e. $\chi^2_\nu \equiv \chi^2/N$ that has mean value 1.

Straight line fitting

- Let's study a few examples fitting a straight line $y = ax + b$. $Y = \frac{y_i - (ax_i + b)}{\sigma_{y_i}}$ should be normally distributed with $\mu = 0$ and $\sigma^2 = 1$.
- Hence: $\chi^2 = \sum_{i=1}^N \left(\frac{y_i - (ax_i + b)}{\sigma_{y_i}} \right)^2$ will follow a χ^2 distribution with $N - 2$ degrees of freedom (one estimates two free parameters from the N datapoints).



```

1 begin
2   # Lest's define a function to fit and a chi2 to minimize
3
4   f(x;a=1,b=0) = a.*x .+ b
5
6   χ2(prs) = sum((f(x,a=prs[1],b=prs[2]) .- y).^2 ./ σ.^2)
7
8
9   fg7 = Figure(size=(1000,400))
10
11
12   # case 1
13   x = [0.96,1.95,2.93,3.98,4.97,6.07,7.02,8.06,8.94,10.05]
14   y = [-3.99,25.48,52.17,49.88,44.745,65.00,74.76,69.23,103.26,100.37]
15   σ = [9.53,10.12,9.46,8.86,9.80,9.56,9.69,9.35,9.97,9.48]
16
17   x0 = [10.0, 5.0]
18   res = optimize(χ2, x0)
19
20   prs1 = Optim.minimizer(res)
21   χ21 = χ2(prs1)
22   c21 = format(χ2(prs1),precision=2)
23   c2r1 = format(χ2(prs1)/(length(x)-2),precision=2)
24   title = latexstring("\chi^2="*c21*",\chi^2_\nu="*c2r1*)
25
26   ax1fg7 = Axis(fg7[1, 1],
27     title = title
28   )
29
30   scatter!(x,y,color=:blue)
31   errorbars!(x,y,σ,color=:blue)
32   lines!(0:12,f(0:12,a=prs1[1],b=prs1[2]),color=:blue)
33
34   xlims!(0,12)
35   ylims!(-20,120)
36
37
38   # case 2
39   x = [0.94,1.89,2.94,3.99,4.96,5.98,7.01,8.09,9.0,10.0]
40   y = [10.08,21.47,43.07,90.58,101.73,137.73,180.6,242.09,289.55,354.8]
41   σ = [8.57,8.9,9.75,11.88,11.19,11.05,10.83,11.09,11.66,10.92]
42
43   x0 = [10.0, 5.0]
44   res = optimize(χ2, x0)
45
46   prs2 = Optim.minimizer(res)

```

```

47  $\chi^2_2 = \chi^2(\text{prs2})$ 
48  $c22 = \text{format}(\chi^2(\text{prs2}), \text{precision}=2)$ 
49  $c2r2 = \text{format}(\chi^2(\text{prs2})/(\text{length}(x)-2), \text{precision}=2)$ 
50  $\text{title} = \text{latexstring}("\chi^2="*\underline{c22}*, "\chi^2_{\nu}="*\underline{c2r2})$ 
51
52  $\text{ax1fg7} = \text{Axis}(\text{fg7}[1, 2],$ 
53      $\text{title} = \underline{\text{title}}$ 
54 )
55
56  $\text{scatter!}(x, y, \text{color}=:blue)$ 
57  $\text{errorbars!}(x, y, \sigma, \text{color}=:blue)$ 
58  $\text{lines!}(0:12, f(0:12, a=\text{prs2}[1], b=\text{prs2}[2]), \text{color}=:blue)$ 
59
60  $\text{xlims!}(0, 12)$ 
61  $\text{ylims!}(0, 400)$ 
62
63
64 # case 3
65  $x = [1.02, 2.0, 3.0, 3.92, 5.02, 5.97, 7.05, 8.03, 9.03, 9.97]$ 
66  $y = [5.34, 24.13, 32.08, 45.38, 56.59, 67.88, 72.01, 85.41, 93.09, 102.65]$ 
67  $\sigma = [8.93, 9.87, 10.14, 9.65, 9.7, 10.13, 9.59, 9.7, 9.39, 9.63]$ 
68
69  $x0 = [10.0, 5.0]$ 
70  $\text{res} = \text{optimize}(\chi^2, x0)$ 
71
72  $\text{prs3} = \text{Optim.minimizer}(\text{res})$ 
73  $\chi^2_3 = \chi^2(\text{prs3})$ 
74  $c23 = \text{format}(\chi^2(\text{prs3}), \text{precision}=2)$ 
75  $c2r3 = \text{format}(\chi^2(\text{prs3})/(\text{length}(x)-2), \text{precision}=2)$ 
76  $\text{title} = \text{latexstring}("\chi^2="*\underline{c23}*, "\chi^2_{\nu}="*\underline{c2r3})$ 
77
78  $\text{ax1fg7} = \text{Axis}(\text{fg7}[1, 3],$ 
79      $\text{title} = \underline{\text{title}}$ 
80 )
81
82  $\text{scatter!}(x, y, \text{color}=:blue)$ 
83  $\text{errorbars!}(x, y, \sigma, \text{color}=:blue)$ 
84  $\text{lines!}(0:12, f(0:12, a=\text{prs3}[1], b=\text{prs3}[2]), \text{color}=:blue)$ 
85
86  $\text{xlims!}(0, 12)$ 
87  $\text{ylims!}(0, 120)$ 
88
89
90  $\text{fg7}$ 
91 end

```

- Inspecting results visually, we might state that:
 - case 1** is an acceptable fit, residuals look randomly distributed and of the "right size" (i.e. if errors are at 1σ , roughly 30% of the points should be farther than 1σ from the fit).
 - case 2** is clearly unacceptable. Residuals look "big" and definitely not randomly distributed.
 - case 3** shows very small residuals, but this is not a signature of a good fit. Typically, this means errors are overestimated (or points are not statistically independent).
- We can formalize part of this visual feeling computing the probability, based on the χ^2 distribution, to have residuals at least as large as, or larger, than measured.
 - This is just the integral of the χ^2 distribution with the given degrees of freedom from the obtained χ^2 value to infinite.

Probability

- Case 1: 0.13

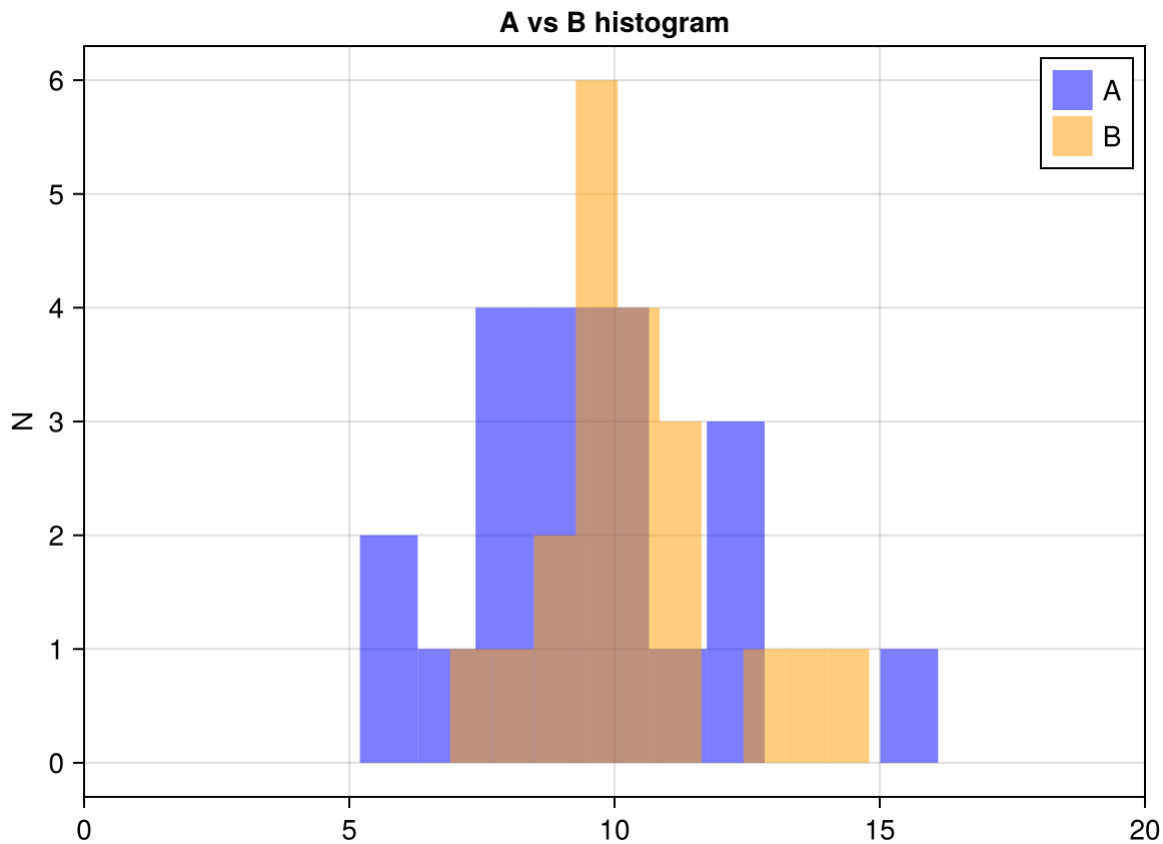
- Case 2: $2.96e - 07$

- Case 3: 0.997

- These results can be read saying that in **case 1** there is about 13% probability to get residuals as large as or larger than measured. In **case 2** this probability is negligible and for **case 3** is very high.
- **Case 1**, depending on the threshold one may define, is likely an acceptable fit. **Case 2** means that the model is highly inadequate, and **case 3** means that residuals are so low to be very suspicious about the input data.

The Kolmogorov-Smirnov (K-S) test

- It is a general test to determine if two datasets "differ" in a statistical sense, i.e. if they can be drawn from the same parent distribution.
- It can also be used to check the consistency of a dataset with a parent distribution.
- It is very general, since there is no assumption about the shape of the population.

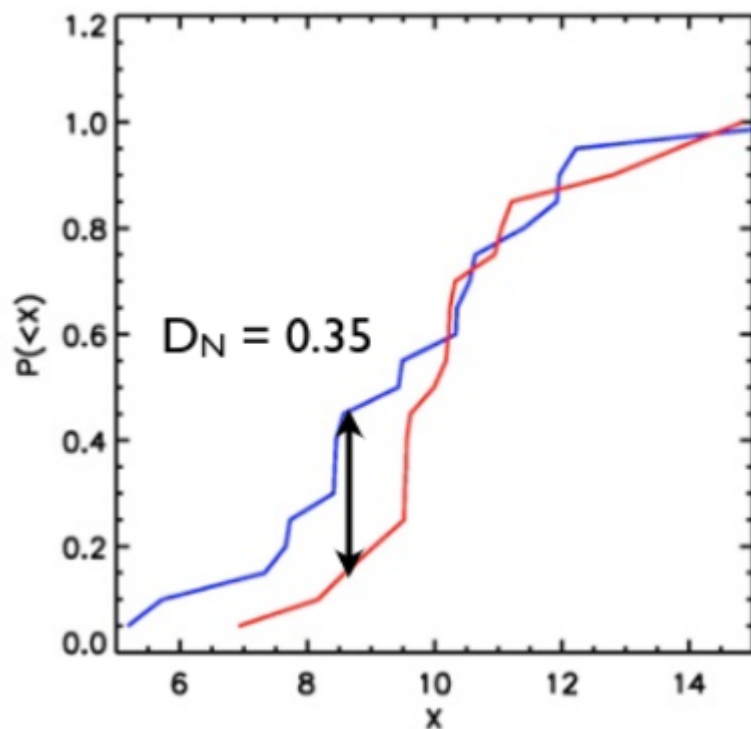


```

1 begin
2   # Let's define two datasets
3
4   A =
5   [9.5,11.4,16.1,8.4,9.4,7.7,10.6,5.7,7.7,10.3,12.0,5.2,8.4,11.9,7.3,12.2,8.6,10.4,10.
6   6,8.5]
7
8   B =
9   [10.2,9.5,10.2,8.6,8.2,10.3,13.8,12.8,11.2,10.0,6.9,10.2,9.0,9.5,10.9,9.6,9.5,11.1,1
10  4.8,9.6]
11
12   fg8 = Figure()
13   ax1fg8 = Axis(fg8[1, 1],
14                 ylabel = "N",
15                 title = "A vs B histogram"
16   )
17
18   hist!(A,color=(:blue,0.5),label="A",bins=10)
19   hist!(B,color=(:orange,0.5),label="B",bins=10)
20
21   xlims!(0,20)
22
23   axislegend()
24
25   fg8
26 end

```

- So, are the two distributions consistent?
- The KS compares the two cumulative distributions and measures the maximum "vertical" distance:



- The KS test p-value measures the probability that a given distance can arise by chance.

Approximate two sample Kolmogorov-Smirnov test

Population details:

parameter of interest:	Supremum of CDF differences
value under h_0 :	0.0
point estimate:	0.3

Test summary:

outcome with 95% confidence:	fail to reject h_0
two-sided p-value:	0.3291

Details:

number of observations:	[20,20]
KS-statistic:	0.9486832980505138

```
1 ApproximateTwoSampleKSTest(A,B)
```

- And we cannot reject the hypothesis that the two datasets are drawn from the same distribution.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [E. Feigelson & G.J. Babu - "Modern Statistical Methods for Astronomy"](#)
- [S.H. Chan - "Introduction to Probability for Data Science"](#)

Credits

This notebook contains material obtained by <https://towardsdatascience.com/fundamental-statistics-7770376593b> and https://dk81.github.io/dkmathstats_site/prob-inverse-cdf.html#:~:text=The%20Inverse%20CDF%20Method%20allows,x%E2%88%92ab%E2%88%92a. and https://dk81.github.io/dkmathstats_site/prob-inverse-cdf.html#:~:text=The%20Inverse%20CDF%20Method%20allows,x%E2%88%92ab%E2%88%92a.

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Introduction	Reminder of Bayesian statistics	Course Summary
html	Introduction	Reminder of Bayesian statistics	Course Summary

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